JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

Affiliated to Dr. Babasaheb Ambedkar Technological University, Lonere

Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in

Education to Eternity

An Autonomous Institute, with NAAC "A" Grade

Basic Science & Humanities Department

2023-24 (Even sem)

VISION	MISSION
To be a Department providing high quality & globally competent knowledge of concurrent technologies in the field of Electronics and Telecommunication."	1. To provide quality teaching learning process through well-developed educational environment and dedicated faculties. 2. To produce competent technocrats of high standards satisfying the needs of all stakeholders.

UNIT TEST EXAMINATION

Course: B. Tech. in CSE, AI, DS

Subject: Basic Civil and Mechanical Engineering

Subject Code: CS2T004, AI2T004, DS2T004

Year/Semester: 2nd Semester (1st Year)

Max Marks: 20 **Date: 02/04/2024**

Duration:- 1 Hr.

Instructions to the Students:

1. **All the questions carry marks as indicate.**
2. **Assume suitable data whenever necessary.**
3. **Illustrate your answer whenever necessary with the help of neat sketches.**

Students should be able to:

1. **Define basic stream of Mechanical & Civil Engineering.**
2. **Explain the concepts of product manufacturing, Energy engineering, design engineering, Automobile engineering, construction technique and civil surveying.**
3. **Apply Basic knowledge of Casting, Machining, Designing, Manufacturing & Civil Construction technique.**
4. **Analyzed the different mechanical system and properties of construction & surveying material.**
5. **Interpret the problem in mechanical system and civil structure.**

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	What are the types of manufacturing processes?	2	1
Q.2	Define Entropy.	1	1
Q.3	Classify heat engine.	3	1
Q.4	Define quality of product.	2	1

SECTION B

1. **All Questions are compulsory.**
2. **Answer in one two three sentence.**

Q.5	How the nuclear fission reaction is performed?	1	2
Q.6	Explain external combustion engine with proper example.	3	2
Q.7	State the types of thermodynamic system.	1	2
Q.8	What is meant by enthalpy.	1	2

SECTION C

Solve any two questions.

Q.9	State the laws of thermodynamics with example.	1	4
Q.10	Explain internal combustion engine with its neat labeled sketch.	3	4
Q.11	Explain steam turbine power plant with neat sketch.	2	4

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